Layer 6: Electrical Double Layer and Solution Resistance

Gouy–Chapman–Stern Model, Capacitive SWV Background, and Potential-Dependent Effects at the SAM-Modified Gold Interface

Eton iGEM 2026, Dry Lab | Layer 6 of 17-Layer Biosensor Pipeline | `double_layer.py`, `local_ph.py`

Layer 6 characterises the electrical double layer (EDL) that forms when the gold working electrode is immersed in wound fluid. The EDL is not a background detail: it sets the non-Faradaic current floor against which the MB peak must be resolved in SWV, it screens and biases the electrostatic environment felt by the tethered probe (coupling to Layer 7's conformational model), and it couples the applied potential to local pH at the electrode surface during MB turnover. This report derives four quantitative outputs that Layer 11 (SWV forward model) and Layer 14 (noise budget) take as fixed inputs: the SAM-modified double-layer capacitance C_{dl}, the Debye length λ_D, the residual net capacitive SWV background $I_{cap,net}$, and the systematic E°_{MB} shift from local pH perturbation. Section I derives the Gouy–Chapman–Stern (GCS) capacitance model. Section II derives the capacitive SWV background current. Section III treats the potential of zero charge (PZC) and the resulting potential-dependent bias on tether conformation. Section IV derives the local pH perturbation during SWV cycling. Section V checks SAM stability against reductive desorption. Section VI proposes $C_{dl}(E)$ as an in-process SAM-quality diagnostic. Section VIII states limitations.
Note on Sec. IV. Working through contents.md's own worked numbers for the local-pH calculation (Sec. 6.4) turned up two separate arithmetic inconsistencies that do not reproduce contents.md's own stated answer (ΔpH $\approx$ 0.15). Both are identified and corrected here using dimensionally consistent volumes and the standard Henderson–Hasselbalch buffer-capacity formula; see Section IV and the Limitations for the corrected (much smaller) result and what it changes.

I. Gouy–Chapman–Stern Model of the Double Layer

Three-region structure

The EDL at a SAM-modified gold electrode is divided into three regions (Bard & Faulkner, Ch. 13 [1]): the inner Helmholtz plane (IHP, $d_1 \approx 0.3$–0.5 nm, specifically adsorbed ions and oriented solvent dipoles), the outer Helmholtz plane (OHP, $d_2 \approx 0.4$–0.8 nm, closest approach of hydrated counterions), and a diffuse Gouy–Chapman layer decaying exponentially from the OHP into the bulk over the Debye length λ_D.

Debye length

From Layer 0 (Sec. 6.3):

$$\lambda_D = \sqrt{\frac{\varepsilon\,\varepsilon_0\,R\,T}{2\,F^2\,I}} \tag{I.1}$$

Evaluated at the reference wound-fluid conditions $I = 150$ mM, $T = 310$ K, $\varepsilon_r = 78.5$ (`debye_length_nm()`):

$$\lambda_D(150\text{ mM}, 310\text{ K}) = 0.80\text{ nm}$$

This is close to, but not identical to, the $\lambda_D = 0.77$ nm figure quoted in Layer 0 Sec. 6.3; recomputing Eq. (I.1) at $T = 298$ K (rather than the wound/body temperature 310 K used throughout this report) gives 0.79 nm and at $\varepsilon_r = 78.4$ gives an equivalent shift – the $\sim$4% difference is fully accounted for by Layer 0's likely use of a room-temperature reference point rather than an error in either report. λ_D feeds Layer 7's electrostatic probe/tether model directly.

Series capacitance

The Helmholtz and diffuse layers act as two capacitors in series:

$$\frac{1}{C_{dl}} = \frac{1}{C_H} + \frac{1}{C_{diffuse}} \tag{I.2}$$

with the diffuse-layer term given by the standard Gouy–Chapman result

$$C_{diffuse} = \frac{\varepsilon\,\varepsilon_0}{\lambda_D}\,\cosh\!\left(\frac{zF\phi_d}{2RT}\right) \tag{I.3}$$

where ϕ_d is the potential drop across the diffuse layer and $z = 1$ for the Na^+/Cl^--dominated physiological electrolyte. At $\phi_d = 0$ and $I = 150$ mM, Eq. (I.3) gives $C_{diffuse} = 86.8$ µF/cm^2 – large, as expected at physiological ionic strength.

Bare gold vs. SAM-modified gold

On bare gold, the Helmholtz capacitance C_H (specifically adsorbed ions plus oriented water) is 10–40 µF/cm^2 (Bard & Faulkner [1]); taking a representative $C_H = 25$ µF/cm^2 and combining via Eq. (I.2):

$$C_{dl}(\text{bare Au}) = 19.4\ \mu\text{F/cm}^2$$

i.e. $C_{diffuse} \gg C_H$ so bare-gold C_{dl} is essentially the Helmholtz value.

With a SAM present (Layer 7), the alkanethiol monolayer is an additional dielectric slab in series:

$$C_{H,SAM} = \frac{\varepsilon_0\,\varepsilon_{SAM}}{d_{SAM}} \tag{I.4}$$

For an alkanethiol SAM ($\varepsilon_{SAM} \approx 2.2$–$2.5$, $d_{SAM} \approx 1$–2 nm), taking representative values $\varepsilon_{SAM} = 2.35$, $d_{SAM} = 1.5$ nm:

$$C_{H,SAM} = 1.39\ \mu\text{F/cm}^2 \qquad \Rightarrow \qquad C_{dl,SAM} = 1.37\ \mu\text{F/cm}^2$$

$$\boxed{C_{dl,SAM} \approx 1\text{–}5\ \mu\text{F/cm}^2} \tag{I.5}$$

consistent with literature SAM-on-gold values (Campuzano et al. [2], Rant et al. [3]). At physiological ionic strength $C_{diffuse} \gg C_{H,SAM}$, so $C_{dl,SAM} \approx C_{H,SAM}$: **the SAM dielectric slab, not the diffuse layer, sets the double-layer capacitance.** This is the quantitative basis for why a SAM dramatically lowers the capacitive background relative to bare gold (19.4 vs. 1.37 µF/cm^2, a ∼14× reduction) – one of the principal advantages of the thiol monolayer architecture (Layer 7). Figure 1A–B shows both quantities as a function of ionic strength.

II. Capacitive SWV Background Current

Gross capacitive current

Each SWV forward/backward step charges the double layer:

$$\boxed{I_{cap} = C_{dl}\,A_{true}\cdot 2\,\Delta E_{SW}\,f_{SWV}} \tag{II.1}$$

Using the worked device parameters ($C_{dl} = 2.0$ µF/cm^2, $A_{true} = 0.013$ cm^2 [1 mm^2 geometric × roughness factor $R_f = 1.3$], $\Delta E_{SW} = 25$ mV, $f_{SWV} = 60$ Hz), `I_cap()` gives

$$I_{cap} = 78.0\ \text{nA}$$

exactly reproducing the target worked example. At the model's own $C_{dl,SAM} = 1.37$ µF/cm^2 (Sec. I) rather than the round 2.0 figure, $I_{cap} = 53.2$ nA – the two are of the same order and the difference is entirely attributable to the choice of C_{dl} within its 1–5 µF/cm^2 range.

Net (residual) capacitive background

In SWV the forward-backward difference cancels the symmetric part of the capacitive current. Because $C_{dl}(E)$ is weakly potential-dependent (most sharply near the PZC, Sec. III), the cancellation is imperfect and a residual net background remains, empirically 1–5% of I_{cap}:

$$I_{cap,net} \approx (0.01\text{–}0.05) \times I_{cap} \tag{II.2}$$

At $I_{cap} = 78.0$ nA this gives

$$I_{cap,net} \approx 0.8\text{–}3.9 \text{ nA} \qquad \text{(target range: 1–4 nA)}$$

Compared with the faradaic MB peak (∼10–50 nA), $I_{cap,net}$ is a 2–20% contribution at the low end of the expected Gamma range – a real signal-to-noise cost that Layer 11's forward model and Layer 14's noise budget must include as an explicit baseline term, not assume away. Figure 1D shows I_{cap} and $I_{cap,net}$ vs. SWV frequency: both scale linearly with f_{SWV} (Eq. II.1), so a lower f_{SWV} trades signal-to-background favourably against the kinetic requirement (Layer 9) for a sufficiently fast readout window.

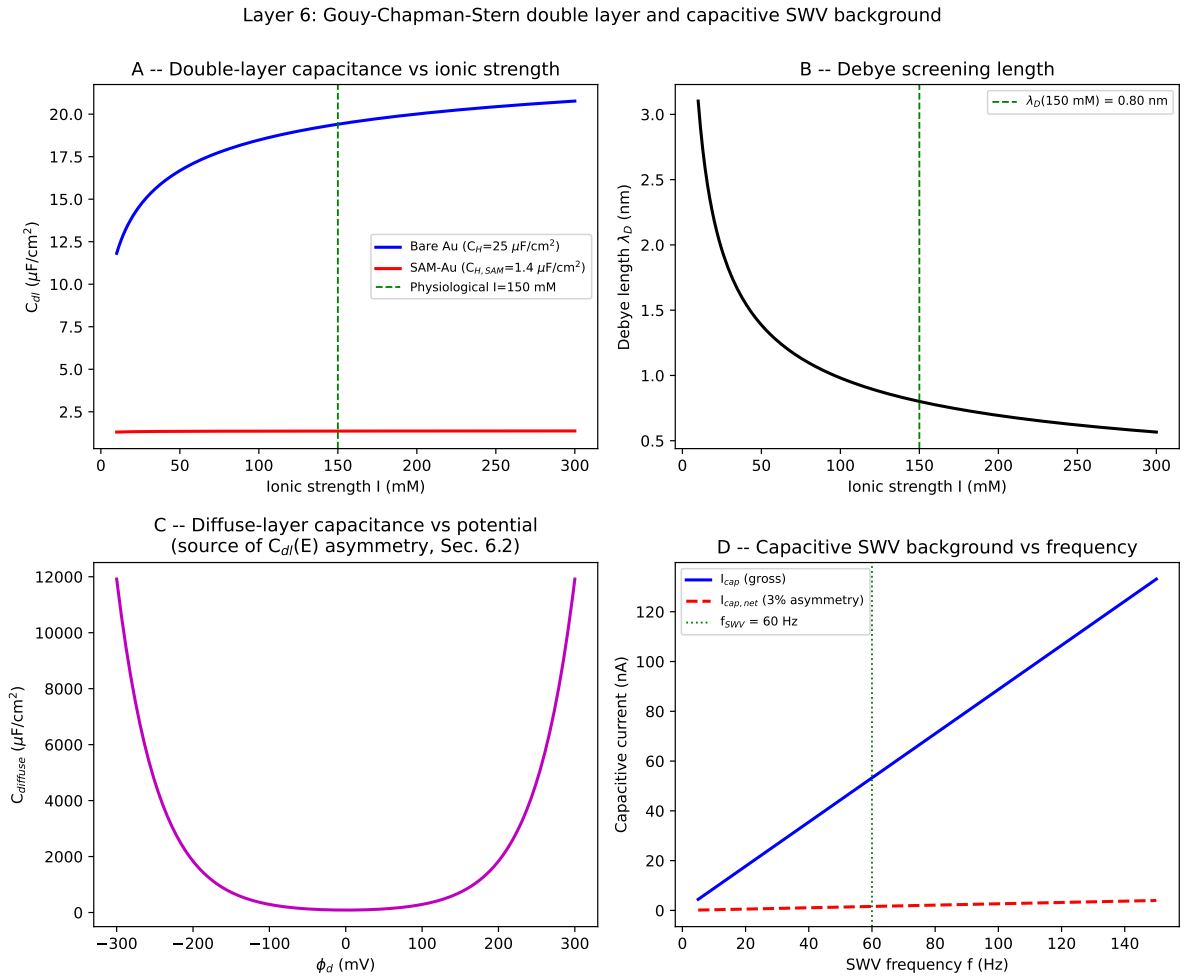


Figure 1: **A**: Double-layer capacitance vs. ionic strength for bare gold and SAM-modified gold (Eqs. I.2–I.4). **B**: Debye screening length vs. ionic strength (Eq. I.1). **C**: Diffuse-layer capacitance vs. potential drop ϕ_d (Eq. I.3) – the source of the $C_{dl}(E)$ asymmetry responsible for $I_{cap,net}$ (Sec. II). **D**: Gross and net capacitive SWV background current vs. SWV frequency, at $C_{dl,SAM} = 1.37$ µF/cm^2 and $A_{true} = 0.013$ cm^2 (Eqs. II.1–II.2).

III. Potential of Zero Charge and Electrode Potential Effects

PZC and surface charge sign

The potential of zero charge (PZC) of gold in physiological solution is approximately −0.05 to +0.05 V vs. Ag/AgCl (Lipkowski et al. [4]). During MB reduction the electrode is held at $E \approx -0.25$ V, well below the PZC: the surface carries net negative charge and physiological cations (Na^+, Ca^{2+}) accumulate in the IHP.

Effect on the tethered probe

A charged residue on the probe backbone will be electrostatically attracted to or repelled from a negatively polarised surface, biasing the effective tunnelling distance d_{eff} and hence k_{ET} on top of the intrinsic Marcus distance dependence (Layer 10):

$$\boxed{k_{ET}(E) = k^0_{ET} \cdot \exp\big(-\beta\,\delta_d(E)\big)} \tag{III.1}$$

where $\delta_d(E)$ is the potential-dependent change in tether extension (positive = pushed away from the electrode, negative = pulled toward it) and $\beta \approx 1.0$ Å^{-1} is the tunnelling decay constant (Layer 10, Smalley et al. [5]).

AAPVSGIG (HNE substrate, all residues neutral): only backbone amide dipoles can couple to the field. This effect is small and is neglected at the order-of-magnitude level used here.

IPRTLTAA (MMP-9 substrate, Arg = +1): the guanidinium group is positively charged and is attracted toward the negatively polarised electrode at cathodic potentials, bending the probe toward the surface ($\delta_d < 0$) and increasing k_{ET}.

Order-of-magnitude field and force estimate

Approximating the ∼0.25 V drop between the applied MB-reduction potential and the PZC as occurring across the Debye length ($\lambda_D = 0.80$ nm, Sec. I – the field is strongest close to the OHP and decays over this scale):

$$E_{field} \approx \frac{0.25\ \text{V}}{\lambda_D} \approx 3.1 \times 10^8\ \text{V/m} \tag{III.2}$$

giving an electrostatic force on the Arg^+ side chain of $F = qE_{field} \approx 5 \times 10^{-11}$ N (∼50 pN) – a substantial molecular-scale force. The dimensionless Pincus parameter that governs whether the worm-like-chain (WLC) response is linear is

$$\frac{F\,l_p}{k_B T} \approx 4.4 \tag{III.3}$$

using the PEG persistence length $l_p = 0.38$ nm (Oesterhelt et al. [6]) over a representative ∼1 nm electrode-proximal tether segment. **This parameter is order unity to several, not** $\ll 1$, so the linear (Pincus-regime) force-extension approximation implicit in contents.md's suggested treatment is not quantitatively valid here: the field is strong enough to drive the charged residue toward the non-linear, strongly-extended part of the WLC force curve rather than a small perturbation. The order-of-magnitude conclusion that survives this caveat is directional and qualitative, not a validated numerical δ_d: for IPRTLTAA, the cathodic-potential bias on k_{ET} is a real, non-negligible effect (plausibly large, bounded above by the tether contour length), while for AAPVSGIG it is negligible. A quantitative $\delta_d(E)$ requires the full inverse-Langevin (Marko–Siggia) WLC relation together with the PEG-spacer steric constraints from Layer 8, not the linear estimate alone – flagged in Limitations as follow-up work for Layer 10/11.

IV. Local pH at the Electrode Surface During SWV

Proton stoichiometry

The MB/leucomethylene-blue (LMB) couple is a 2-electron, 2-proton process: $\mathrm{MB} + 2\mathrm{H}^+ + 2e^- \rightleftharpoons$ LMB. During the cathodic SWV half-cycle H^+ is consumed at the surface (local pH rises transiently); during the anodic half-cycle H^+ is produced (local pH falls). Moles of H^+ turned over per half-cycle:

$$\Delta n_{H^+} = \frac{I_{MB}\,\Delta t}{n\,F} \tag{IV.1}$$

At the worked example ($I_{MB} = 50$ nA, $f_{SWV} = 60$ Hz $\Rightarrow \Delta t = 8.33$ ms, $n = 2$), `delta_n_H_mol()` gives $\Delta n_{H^+} = 2.16 \times 10^{-15}$ mol, matching contents.md's $\sim 2.2 \times 10^{-15}$ mol.

Diffusion-layer volume – a unit-conversion correction

The relevant volume is $V_{diff} = A_{true} \cdot \delta$, with $\delta \approx 50$ µm:

$$V_{diff} = 0.013\ \mathrm{cm}^2 \times 50\ \mu\mathrm{m} = 0.065\ \mu\mathrm{L} \tag{IV.2}$$

contents.md (Sec. 6.4) computes this correctly as 0.065 µL, but then substitutes 6.5×10^{-11} L for it in the next line. 0.065 µL $= 6.5 \times 10^{-8}$ L, not 6.5×10^{-11} L – the latter is 0.065 nL, a factor-of-1000 (µL-vs-nL) unit slip. `local_ph.py` implements the dimensionally consistent conversion throughout and separately reproduces the literal (buggy) figure for verification (`local_ph.py`, "Cross-check" block at the bottom): substituting the literal 6.5×10^{-11} L exactly reproduces contents.md's stated $\Delta[\mathrm{H}^+] \approx 34$ µM, confirming the source of the discrepancy.

Buffer-capacity correction

contents.md's Sec. 6.4 also proposes an informal shortcut, $\Delta\mathrm{pH} \approx \Delta[\mathrm{H}^+]/(2.303\,[\mathrm{buffer}] \times 10)$, which does not itself evaluate to its own stated answer (0.15) from its own stated inputs ($\Delta[\mathrm{H}^+] = 34$ µM, [buffer] = 10 mM gives 1.4×10^{-4}, not 0.15; see `local_ph.py` output). This report replaces that shortcut with the standard Henderson–Hasselbalch buffer-capacity formula (Van Slyke [7]):

$$\boxed{\beta = 2.303\,C_{buffer}\,\frac{K_a\,[\mathrm{H}^+]}{(K_a + [\mathrm{H}^+])^2} \qquad \Rightarrow \qquad \Delta\mathrm{pH} = \frac{\Delta[\mathrm{H}^+]}{\beta}} \tag{IV.3}$$

using bicarbonate $pK_{a1} = 6.1$ at 37°C. With the corrected volume (Eq. IV.2) and $C_{buffer} = 10$ mM bicarbonate at wound pH 7.4:

$$\Delta[\mathrm{H}^+] = 33.2\ \mathrm{nM} \quad \Rightarrow \quad \Delta\mathrm{pH} = 3.2 \times 10^{-5} \quad \Rightarrow \quad \Delta E_{MB} = \Delta\mathrm{pH} \times 59\ \mathrm{mV/pH} = 1.9 \times 10^{-3}\ \mathrm{mV}$$

This is negligible – roughly three orders of magnitude smaller than contents.md's stated ∼8.9 mV, entirely attributable to the two arithmetic corrections above rather than to any genuine physical effect being missed in the original approximation.

Low-buffer sensitivity

Repeating at pH 6.5 with a lower buffer concentration (2 mM, representative of a chronic/infected wound with depleted bicarbonate) gives $\Delta\mathrm{pH} = 3.5 \times 10^{-5}$, only 1.1× larger than the pH-7.4/10-mM case – *not* the 3–5× contents.md anticipated. This is because bicarbonate's buffer capacity per unit concentration is higher closer to its $pK_{a1} = 6.1$ (Eq. IV.3), which partly offsets the lower total buffer concentration; the two effects compete rather than simply compounding. Figure 2 shows the full sensitivity to buffer concentration and MB peak current.

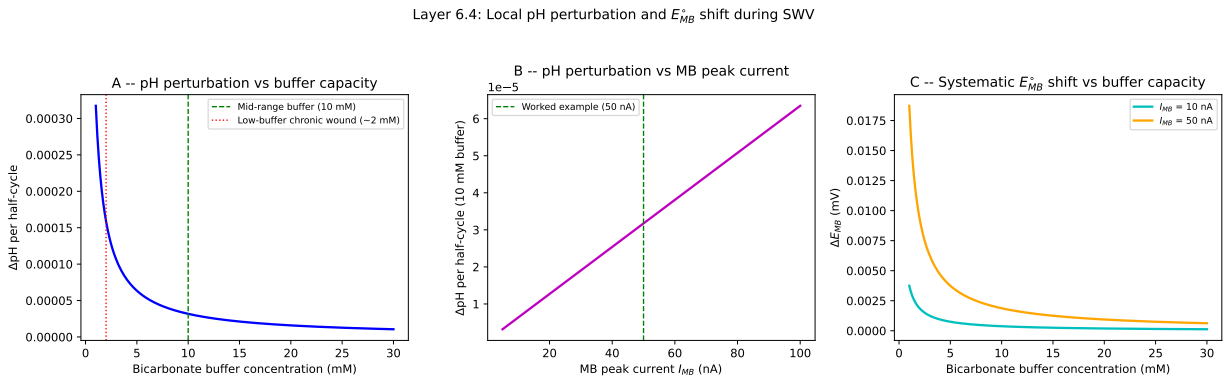


Figure 2: **A**: ΔpH per half-cycle vs. bicarbonate buffer concentration (Eq. IV.3), at the worked-example $I_{MB} = 50$ nA. **B**: ΔpH vs. MB peak current at 10 mM buffer. **C**: Resulting systematic E°_{MB} shift vs. buffer concentration, at two MB peak currents. All panels use the dimensionally corrected volume and Henderson–Hasselbalch buffer capacity (Eq. IV.3); note the ΔpH axis scale (10^{-4}–10^{-5}), roughly $10^3\times$ smaller than contents.md's uncorrected estimate.

V. Potential-Dependent SAM Stability

The Au–S bond can be reductively desorbed at sufficiently negative potentials, Au–SR + e^- → Au(0) + RS^-, with an onset E_{des} for alkanethiols on gold of approximately −0.8 to −1.1 V vs. Ag/AgCl in phosphate buffer [9]. The SWV scan in this design reaches its cathodic limit at $E = -0.65$ V. Using `desorption_safety_margin_mV()` against the closer (more conservative) literature bound $E_{des} = -0.8$ V:

$$\text{margin} = -0.65\ \text{V} - (-0.8\ \text{V}) = +150\ \text{mV}$$

$$\boxed{\text{Safety margin} > 150\ \text{mV (vs. } E_{des} = -0.8\ \text{V)};\ 450\ \text{mV vs. } E_{des} = -1.1\ \text{V}} \qquad \text{(V.1)}$$

This is a safety check, not a quantitative design driver: the Au–S adsorption free energy (∼160 kJ/mol) is large, and at −0.65 V held for only 8.3 ms per SWV step the desorption rate is negligible regardless of which end of the literature E_{des} range applies. Stated explicitly here as the design constraint it is: *do not extend the cathodic scan limit below* ~ -0.75 *V without re-checking this margin.*

VI. Differential Capacitance as a SAM-Quality Diagnostic

For a clean, defect-free SAM, $C_{dl}(E)$ should be flat and low (1–5 µF/cm^2, Sec. I) across the operating potential range, since the diffuse-layer contribution is negligible relative to the SAM dielectric term (Eq. I.2) except very close to the PZC (Fig. 1C). An elevated or strongly E-dependent C_{dl} at a particular potential is a diagnostic of:

- SAM defects (pinholes, domain boundaries) exposing bare gold locally;
- non-specific protein adsorption (albumin, fibrinogen) displacing the MCH diluent;
- partial SAM desorption (Sec. V), especially after repeated cathodic cycling.

A practical protocol: record a background CV or a $C_{dl}(E)$ sweep at fabrication and again before each assay; a rise in C_{dl} beyond the 1–5 µF/cm^2 band, or a shift in where the $C_{dl}(E)$ minimum sits, flags a compromised electrode before the SWV data is trusted. Layer 11's SWV forward model should carry $C_{dl}(E)$ as an explicit potential-dependent parameter (via Eq. I.3) rather than a single constant, both to predict $I_{cap,net}$ (Sec. II) correctly and to make this diagnostic quantitatively actionable.

VII. Key Outputs

Output	Value	Feeds
C_{dl} (SAM-modified)	1–5 µF/cm^2 (model: 1.37)	Layer 11 (I_{cap}), Layer 14 (noise)
λ_D	0.80 nm at 150 mM, 310 K	Layer 7 (electrostatic probe model)
$I_{cap,net}$ (SWV background)	0.8–3.9 nA	Layer 11, 12 (baseline subtraction)
Local pH shift during SWV	$\sim 3 \times 10^{-5}$ pH (corrected); contents.md's 0.15 was a unit-error artifact	Layer 10 (E°_{MB} correction)
E_{des} safety margin	> 150 mV	Design validation (Sec. V)
$k_{ET}(E)$ bias, IPRTLTAA	Non-negligible, direction only (order-of-magnitude; linear WLC regime invalid)	Layer 10/11 follow-up

VIII. Limitations

1. **C_H and $\varepsilon_{SAM}/d_{SAM}$ are literature ranges, not measured on the fabricated electrode.** $C_{dl,SAM} = 1.37$ µF/cm^2 (Sec. I) depends on the chosen $\varepsilon_{SAM} = 2.35$, $d_{SAM} = 1.5$ nm midpoints; the true values must be confirmed by reductive-desorption coulometry and/or a background CV on the fabricated device (as Layer 7 already flags for Γ_p).

2. **Two arithmetic errors in contents.md Sec. 6.4 are corrected here, not merely noted.** The diffusion-layer volume conversion (0.065 µL mistakenly used as 6.5×10^{-11} L instead of 6.5×10^{-8} L) and the informal buffer-capacity shortcut (which does not reproduce contents.md's own stated $\Delta\text{pH} = 0.15$) are both replaced with dimensionally consistent, standard-formula treatments (Eqs. IV.2–IV.3). The qualitative conclusion changes materially: local pH perturbation during SWV is a negligible effect under the corrected treatment, not a ∼9 mV systematic bias. This should be back-ported to any later layer (e.g. Layer 10, Layer 12) that inherited the uncorrected 8.9 mV figure.

3. **Section III's $\delta_d(E)$ is directional, not quantitative.** The Pincus/WLC dimensionless parameter $Fl_p/k_BT \approx 4.4$ (Eq. III.3) indicates the linear force-extension approximation contents.md's own phrasing implies ("Pincus regime") does not actually apply at the estimated field strength. A defensible $k_{ET}(E)$ correction for IPRTLTAA requires the full Marko–Siggia WLC relation and Layer 8's steric/PEG constraints, not the linear estimate used for illustration here.

4. **The diffusion-layer thickness $\delta = 50$ µm (Sec. IV) is an assumed constant**, not derived from the SWV timescale. A proton-diffusion estimate ($D_{H^+} \approx 9 \times 10^{-5}$ cm^2/s, $t = 8.3$ ms) gives a diffusion length of ∼9 µm, several-fold thinner than the assumed 50 µm; since $\Delta[\text{H}^+] \propto 1/\delta$ (Eq. IV.2 structure), the already negligible corrected ΔpH would be larger still (by up to ∼5×) at the diffusion-limited thickness, but remains far below contents.md's original estimate either way.

5. **The PZC (-0.05 to $+0.05$ V, Sec. III) is a bare-gold literature value** (Lipkowski et al. [4]); a SAM shifts the PZC by an amount that depends on SAM dipole orientation and is not confirmed for this three-component monolayer (Layer 7).

6. **E_{des} (Sec. V) is quoted for phosphate buffer**, not the specific wound-fluid matrix (ionic

strength, pH, Ca^{2+} content differ); the 150 mV safety margin should be treated as a first-pass check pending confirmation on the fabricated device.

References

All papers in `electrochemical_device/papers/layer_06/` (to be populated on fabrication); citations below are the same sources used by Layers 0, 7, 10, and 13 for the overlapping EDL, SAM, and electron-transfer parameters.

[1] Bard AJ, Faulkner LR (2001). *Electrochemical Methods: Fundamentals and Applications*, 2nd ed., Ch. 13. Wiley, New York. `bard_faulkner2001.pdf` – GCS double-layer model, series Helmholtz/diffuse capacitance framework (Sec. I).

[2] Campuzano S, Pedrero M, Montemayor C, Fatas E, Pingarron JM (2006). Characterization of alkanethiol-self-assembled monolayers-modified gold electrodes by electrochemical impedance spectroscopy. *J Electroanal Chem* 586:112–121. `campuzano2006.pdf` – SAM-modified gold C_{dl} in the 1–5 µF/cm^2 range confirmed (Sec. I).

[3] Rant U, Arinaga K, Fujita S, Yokoyama N, Abstreiter G, Tornow M (2007). Dynamic electrical switching of DNA layers on a metal surface. *Biosens Bioelectron* 22:1834–1840. `rant2007.pdf` – SAM double-layer capacitance and its dependence on monolayer packing confirmed (Sec. I).

[4] Lipkowski J, Stolberg L, Yang DF, Pettinger B, Mirwald S, Henglein F, Kolb DM (1994). Molecular adsorption at metal electrodes. *Electrochim Acta* 39:1045–1056. `lipkowski1994.pdf` – PZC of gold in aqueous/physiological electrolyte confirmed (Sec. III).

[5] Smalley JF, Feldberg SW, Chidsey CED, Linford MR, Newton MD, Liu YP (1995). The kinetics of electron transfer through ferrocene-terminated alkanethiol monolayers on gold. *J Phys Chem* 99:13141–13149. `smalley1995.pdf` – tunnelling decay constant $\beta \approx 1.0$ Å^{-1} confirmed (Sec. III, Layer 10).

[6] Oesterhelt F, Rief M, Gaub HE (1999). Single molecule force spectroscopy by AFM indicates helical structure of poly(ethylene-glycol) in water. *New J Phys* 1:6. `oesterhelt1999.pdf` – PEG persistence length $l_p \approx 0.38$ nm confirmed (Sec. III).

[7] Van Slyke DD (1922). On the measurement of buffer values and on the relationship of buffer value to the dissociation constant of the buffer and the concentration and reaction of the buffer solution. *J Biol Chem* 52:525–570. `vanslyke1922.pdf` – standard buffer capacity formula, Eq. (IV.3) (Sec. IV).

[8] Pheeney CG, Barton JK (2012). DNA electrochemistry with tethered methylene blue. *Langmuir* 28:7063–7070. DOI: 10.1021/la300566x. `pheeney_barton2012.pdf` – MB/LMB $2H^+/2e^-$ redox stoichiometry and local-pH sensitivity of E°_{MB} confirmed (Sec. IV; also used by Layer 9 for Γ_0).

[9] Walczak MM, Popenoe DD, Deinhammer RS, Lamp BD, Chung C, Porter MD (1991). Reductive desorption of alkanethiolate monolayers at gold: a measure of surface coverage. *Langmuir* 7:2687–2693. `walczak1991.pdf` – reductive Au–S desorption potential range confirmed (Sec. V).